Research Paper Segmentation-Driven Analytics for Churn Mitigation: A Data-Centric Approach to European Banking Retention

Abstract

- **The Problem:** Customer churn as a primary threat to retail banking stability.
- **The Approach:** A multidimensional analysis utilizing statistical segmentation (geographic, demographic, financial) to identify high-risk cohorts.
- **The Outcome:** Evidence-based retention strategies that prioritize high-value segments, demonstrated through the correlation between product engagement and attrition.

Introduction

- **Banking Context:** Define the European retail banking landscape (high competition, cost of acquisition).
- **Research Motivation:** Explain why traditional "average churn" metrics are insufficient (masking segment-specific risks).
- **Objective:** To move from reactive churn reporting to proactive, segmentation-based customer retention.

Methodology

- **Data Preparation:** * Briefly describe the dataset cleaning process (handling categorical variables, normalization of CreditScore and Balance).
    - Feature engineering: Mention the creation of the **Balance-to-Salary Ratio** and **Product Intensity Index** as proprietary metrics.
- **Analytical Framework:**
    - Segmentation Logic: Define the criteria for geographic (France, Spain, Germany) and demographic grouping.
    - Statistical Methods: Mention the use of correlation analysis to determine feature importance regarding Exited (churn).

Data Exploration & Empirical Findings

- **Visualizing Trends:** Discuss the distributions identified (e.g., "Churn rate is disproportionately high in Germany vs. France/Spain").
- **High-Value Risk:** Present findings on high-balance customers who exited.

- **Key Dependencies:**
    - *The "Product Sweet Spot":* Describe the relationship between NumOfProducts and retention.
    - *Age & Engagement Interaction:* Report on which age brackets show the highest volatility.

Discussion & Strategic Recommendations

- **Segment-Specific Retention:**
    - *Example:* If analysis shows churn is highest in "Low-Product, Mid-Age" customers, recommend a cross-sell strategy.
    - *Example:* If churn is high for high-balance users, suggest personalized account management interventions.
- **Operationalizing Insights:** Propose how these findings feed into the CRM (Customer Relationship Management) system to trigger alerts for relationship managers when a customer hits "at-risk" thresholds.

Research Methodology

The analysis employs a quantitative approach to extract behavioural patterns from the provided banking dataset. The methodology is divided into three distinct phases:

- **Data Preparation & Feature Engineering:** To ensure analytical depth, raw data underwent cleaning to handle missing values and encoding. We engineered two primary features:
    - *Balance-to-Salary Ratio:* Calculated as Balance/EstimatedSalary, identifying high-liquidity customers who may have higher exit stakes.
    - *Product Intensity:* Calculated as NumOfProducts/Tenur, identifying customers with rapid product adoption versus long-term stable users.
- **Statistical Segmentation:** Customers were partitioned into cohorts based on geographic location (France, Germany, Spain) and age-based demographics (Gen Z, Millennial, Gen X, Boomer).
- **Analytical Framework:** We applied Chi-Square tests to validate the significance of categorical variables (e.g., IsActiveMember, Geography) against the target variable Exited. Correlation matrices were generated to identify multi-collinearity between financial variables and churn probability.

Executive Summary: Strategic Retention & Churn Mitigation Customer churn presents a quantifiable risk to our revenue stability, particularly within the German market segment. Our analysis indicates that the overall churn rate is disproportionately driven by customers with high account balances who have reduced their product usage.